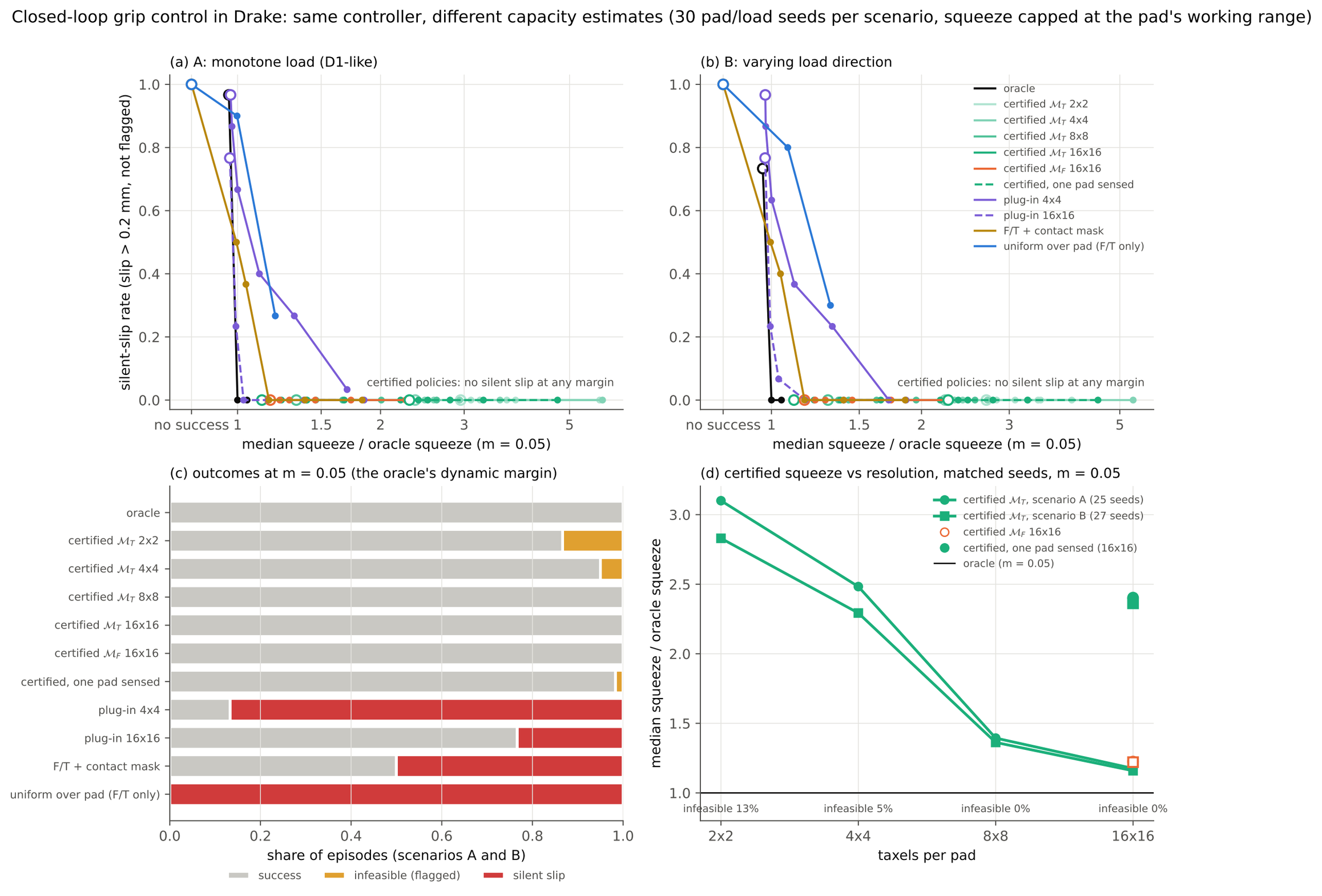
Closed-loop grip control in Drake: same controller, different capacity estimates (30 pad/load seeds per scenario, squeeze capped at the pad's working range)
(a) A: monotone load (D1-like)
(b) B: varying load direction
silent-slip rate (slip > 0.2 mm, not flagged)
median squeeze / oracle squeeze (m = 0.05)
no success
certified policies: no silent slip at any margin
oracle
certified $\mathcal{M}_T$ 2x2
certified $\mathcal{M}_T$ 4x4
certified $\mathcal{M}_T$ 8x8
certified $\mathcal{M}_T$ 16x16
certified $\mathcal{M}_F$ 16x16
certified, one pad sensed
plug-in 4x4
plug-in 16x16
F/T + contact mask
uniform over pad (F/T only)
(c) outcomes at m = 0.05 (the oracle's dynamic margin)
share of episodes (scenarios A and B)
success
infeasible (flagged)
silent slip
(d) certified squeeze vs resolution, matched seeds, m = 0.05
median squeeze / oracle squeeze
taxels per pad
certified $\mathcal{M}_T$, scenario A (25 seeds)
certified $\mathcal{M}_T$, scenario B (27 seeds)
certified $\mathcal{M}_F$ 16x16
certified, one pad sensed (16x16)
oracle (m = 0.05)
infeasible 13%
infeasible 5%
infeasible 0%
infeasible 0%
2x2
4x4
8x8
16x16